

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

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Profile

Senior Data Scientist at Nasdaq; Computer Science Ph.D. Combines financial ML, AWS deployment and generative vision. First-author **IEEE TNNLS/TVCG** papers; **NeurIPS** research integrated into **Hugging Face PEFT**.

Technical Skills

Core: Python, PyTorch, SQL, Git, Bash, C++, CUDA; experimental design and feature engineering.

ML & GenAI: Latent diffusion, Transformers, multimodal AI, PEFT/LoRA, Hugging Face, RAG, agentic tools.

ML systems: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, SLURM.

Professional Experience

Nasdaq

St. John's, NL, Canada

Senior Data Scientist, AI/Analytics

Mar. 2025 – Present

- Designed cross-scale ensemble feature selection for **cheque-fraud ML (CFML)**, removing **77.2% of input features** while retaining competitive performance in evaluations.
- Integrated an **existing cross-team ML UI/backend prototype** into a **SageMaker workspace**; configured networking/IAM and **PostgreSQL user-state persistence**; deployed **FastAPI on AWS Lambda using Mangum**.
- Built **SageMaker Job Submitter** for script/dependency packaging across **Processing, Training, Hyperparameter Tuning, and SageMaker Pipelines**; cut developer job configuration from **7 days to 1 day (about 86% less)**.
- Designed **feature engineering with PyTorch/CUDA-accelerated feature-label AUROC screening**; used **Amazon Bedrock** to assess candidate features' semantic plausibility.
- Developed **PagerDuty RAG assistant v2**, now **used internally and recognized in a company shout-out**; retrieves incident context and troubleshooting guidance for on-call engineers.
- Mentored **cheque OCR** research on perceptually aligned quality metrics, culminating in a student presentation; guided **tabular-data generation** leading to a **StDDPM manuscript submission**.
- Mentored a **work-term student on an input payload checker** (from Jun. 2026) and a **Mitacs student on GNN-based crime-ring detection** (from Sep. 2026).

AI Algorithm Intern (two placements)

May 2024 – Jan. 2025

- Proposed **TextDoctor**: structure-pyramid prediction and patch-pyramid diffusion for document/cheque inpainting; **8K on a 24GB RTX 3090**; **first-author manuscript submitted to EAAI**.
- Evaluated **7 document datasets** without dataset-specific fine-tuning. On FUNSD, TextDoctor (GSDM) achieved **55.38% PaddleOCR word accuracy vs. 45.24% for DocDiff (+10.14 percentage points)**.
- Designed **PLAID**: patch Gaussian discriminant modeling **without backbone fine-tuning**; **8 biometric benchmarks, 16 backbones. 8-shot ResNet-152 AUROC: CEDAR 99.42%, GPDS-150 98.42%**.
- First-author IEEE TIFS submission**; major revision submitted Apr. 2026, awaiting second-round decision. Presented both placements: earlier talk about **40–60 attendees**; final talk **39 cross-functional attendees**.

Beaufort Solutions

St. John's, NL, Canada

AI Intern

Jul. 2022 – Nov. 2022

- Led method design of **TEG**, a CLIP-based **few-shot adaptation** method for photo-book curation; introduced a text-embedding-guided classifier and auxiliary classification loss to improve learning from limited labels.
- Led collection of **Theme25: 35,655 images across 25 themes**; evaluated on Theme25, CIFAR100, and ImageNet. Research published in **Journal of Electronic Imaging (2024)**.

Zhongkong Hanlian

Hangzhou, China

AI Algorithm Intern

Dec. 2017 – Aug. 2018

- Designed **ResNet–DenseNet fusion** for vehicle-logo recognition; built Java annotation tools and a training/evaluation dataset; **deployed with Caffe for real-time inference at community security gates**.

Selected Research Publications

UltraDiffEdit – High-Resolution Generative Editing [Code] First author; IEEE TNNLS 2026

- Developed multi-patch encoding and global-local diffusion sampling for **tuning-free 8K editing on one RTX 3090**; tested SDXL/ControlNet; built **three benchmarks (1,188 images)**.
- **UHRSEdit (4K–8K) crop-FID 7.45 vs. 18.76 for SDXL+bicubic (60.3% lower)**.

FACEMUG – Multimodal Local Facial Editing First author; IEEE TVCG 2024

- Designed multimodal feature fusion and self-supervised latent warping for **six input modalities**: text, sketches, semantic maps, color maps, exemplars, and attribute labels.
- Received **77.88% preference over Unite&Conquer** in a **52-participant CelebA-HQ study** with sketch/semantic/text guidance; preserved unedited regions during local editing.

AdaMSS – Parameter-Efficient Adaptation [Code] Co-author; NeurIPS 2025

- Co-developed adaptive multi-subspace tuning; ViT-Large averaged **90.13% vs. LoRA's 85.40% (+4.73 percentage points)** on **7 tasks** with **15.4% of LoRA's trainable parameters**. In **Hugging Face PEFT**.

DocPure – Unified Document Restoration [Code] Co-first author; IEEE TCSVT 2026

- Co-developed degradation-aware structure extraction and wavelet modulation; tested **one prompt-free model on 12 datasets** for deblurring, denoising, compression artifacts, and deshadowing.
- On TDD deblurring, achieved **32.55 dB PSNR vs. DocRes's 28.34 dB (+4.21 dB)** and **61.27% PaddleOCR word accuracy vs. 55.66% (+5.61 percentage points)**.

GRIG – Data-Efficient Image Inpainting [Code] First author; CVM 2025

- Designed CNN–Transformer residual inpainting with forgery-patch training; **10 few-shot datasets (18–750 training images)**, **3.70–26.48% lower FID vs. next-best baselines at 50–60% masking**.

Visual Style Prompt Learning [Code] First author; Pattern Recognition 2025

- Developed latent-diffusion style prompts and style-modulated aggregation: **four denoising steps, 12.3 FPS at 512 × 512 (RTX 3080 Ti)**; **WebPhoto-Test FID 74.25 vs. CodeFormer's 83.17 (10.7% lower)**.
- Evaluated downstream tasks on **CelebA-Test: FAN landmark NME 6.08% to 2.43%**; **HSEmotion agreement with clean-image predictions 79.73% to 86.03%**, using clean-image model outputs as references.

Full publication record: Google Scholar and publication portfolio.

AI Engineering Prototypes & Deployment

Local Agentic Code Editor Nasdaq internal prototype; 2026

- Built a working local prototype connecting **model APIs to agentic routing across code generation, review and shell tools**; **diagnoses bugs, applies fixes and iterates on execution feedback**.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada 2021–2025

M.Eng., Computer Software and Theory – Wenzhou University, China 2018–2021

B.Eng., Digital Media Technology – Communication University of Zhejiang, China 2014–2018

Selected Recognition & Service

Outstanding Research Award (Ph.D.), Memorial University, 2023; four granted Chinese invention patents.

Mentored **12 students** in generative vision and applied ML, including restoration and recognition projects.

Teaching assistant in Data Structures and Algorithms (2021) and Introduction to Machine Learning (2024).

Invited reviewer for **IEEE TPAMI, TIP, TMM, TCSVT, JBHI and SPL**; **Pattern Recognition**, EAAI, JVC, Knowledge-Based Systems, Neurocomputing, Applied Soft Computing, Scientific Reports, Displays; ECCV 2026.

Presented feature-selection research to **60+ attendees** (2026); invited speaker, IEEE Newfoundland and Labrador Section AGM (2026); guest lecturer on deep generative models, Memorial University (2025).